nature energy

Article

https://doi.org/10.1038/s41560-023-01442-1

Enhancing the efficiency and longevity of inverted perovskite solar cells with antimony-doped tin oxides

Received: 17 May 2023

Accepted: 14 December 2023

Published online: 12 January 2024

Check for updates

Jia Li[1,2,6], Haoming Liang[1,2,6], Chuanxiao Xiao[3,4], Xiangkun Jia[1,2], Renjun Guo[1,2], Jinxi Chen[1,2], Xiao Guo[1,2], Ran Luo[1,2], Xi Wang[1,2], Minghui Li[3], Michael Rossier[5], Alina Hauser[5], Flavio Linardi[5], Ezra Alvianto[1,2], Shunchang Liu[1,2], Jiangang Feng[1,2] & Yi Hou[1,2] ✉

Inverted perovskite solar cells possess great potential for single or multi-junction photovoltaics. However, energy and charge losses at the interfaces limit their performance. Here we introduce p-type antimony-doped tin oxides (ATO_x) combined with a self-assembled monolayer molecule as an interlayer between the perovskite and hole-transporting layers (HTL) in inverted solar cells. ATO_x increases the chemical stability of the interface; we show that the redox reaction that commonly took place at the NiO_x/perovskite interface is negligible at the ATO_x/perovskite interface. We demonstrate that ATO_x suppresses non-radiative recombination in the perovskite layer and enhances the depletion at the perovskite/HTL interface for efficient charge extraction. Owing to these combined improvements, we achieve inverted perovskite solar cells with a maximum efficiency of 25.7% (certified steady-state efficiency of 24.8%) for an area of 0.05 cm^2, retained under maximum power point tracking over 500 h and 24.6% (certified steady-state efficiency of 24.0%) for an area of 1 cm^2.

Through extensive worldwide efforts, the power conversion efficiencies (PCEs) of perovskite solar cells (PSCs) with normal (n–i–p) device configuration have increased substantially from 3.8% to 25.7% in just a decade[1,2]. Although the efficiencies of inverted (p–i–n) PSCs still lag behind[3,4], this configuration has become more appealing for single-junction and tandem solar cells as it shows numerous advantages, such as long operating lifetimes, scalability, low cost and compatibility with silicon in tandem solar cells[5–9].

Open-circuit voltage (V_{oc}) and fill factor (FF) are two crucial factors that currently limit the efficiency of inverted PSCs. To reduce the charge recombination rate in inverted PSCs, significant efforts have been devoted to improving the quality of perovskite film and optimizing the perovskite/transport layer interfaces[4,10,11]. At the perovskite/electron-transporting layer (ETL) interface, two-dimensional perovskites and insulating organic molecules have been widely and successfully implemented into state-of-the-art inverted PSCs[10,12,13]. For example, it has been reported that 1,3-propane-diammonium iodide can tune the Fermi level of perovskite film to optimize the energy alignment and the charge extraction. Additionally, 1,3-propane-diammonium iodide modifies the perovskite/C_{60} interface to reduce the non-radiative recombination and improve device voltage[14].

[1]Department of Chemical and Biomolecular Engineering, National University of Singapore, Singapore, Singapore. [2]Solar Energy Research Institute of Singapore (SERIS), National University of Singapore, Singapore, Singapore. [3]Ningbo Institute of Materials Technology and Engineering, Chinese Academy of Sciences, Ningbo City, China. [4]Ningbo New Materials Testing and Evaluation Center Co., Ningbo City, China. [5]Avantama AG, Stäfa, Switzerland. [6]These authors contributed equally: Jia Li, Haoming Liang. ✉e-mail: yi.hou@nus.edu.sg

https://doi.org/10.1038/s41560-023-01442-1

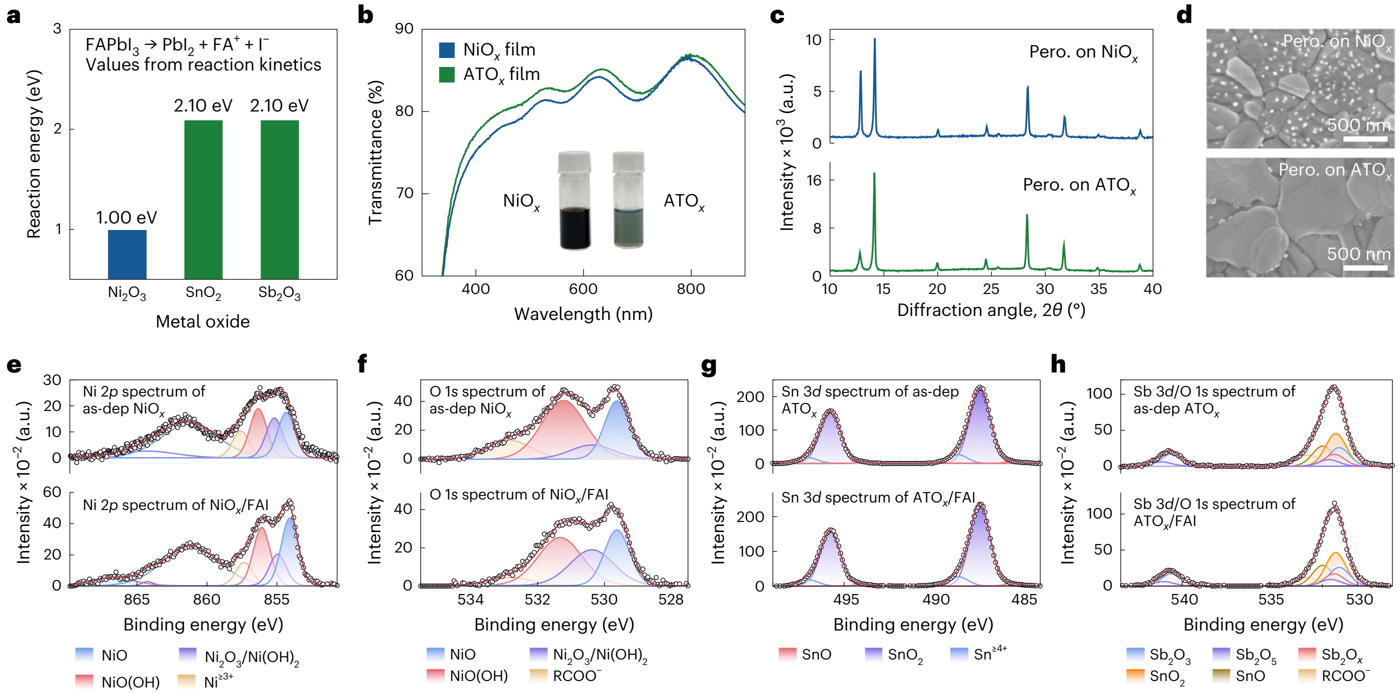

Fig. 1 | Characterization of metal oxide and its interface reaction with perovskite. a, DFT calculated the reaction energy of the redox reaction among $FAPbI_3$ and Ni^{3+}, Sn^{4+} and Sb^{3+}. **b**, UV–vis transmittance spectra of ATO_x and NiO_x films with substrates. The inset shows pictures of ATO_x and NiO_x suspension. **c**,**d**, XRD spectra (**c**) and SEM images (**d**) of perovskite on ATO_x and perovskite on NiO_x with Me-4PACz. **e**,**f**, Ni 2*p* and O 1*s* XPS spectra and fittings of NiO_x and NiO_x/Me-4PACz/Formamidinium Iodide (FAI). **g**,**h**, Sn 3*d*, Sb 3*d* and O 1*s* XPS spectra and fittings of ATO_x and ATO_x/Me-4PACz/FAI. For the XPS spectra, the deconvolution of the components are achieved with the Gaussian–Lorentzian function. The hollow dots illustrate the raw profiles directly from the measurements, and the red lines on top of the hollow dots are constructed from the deconvoluted profiles by summing up all the individual deconvolution components. The close shape of the hollow dots profiles and the red line profiles indicates good fit of the XPS spectra.

At the perovskite/hole-transporting layer (HTL) side, various organic and inorganic HTL materials have been studied, such as poly(bis(4-phenyl)(2,4,6-trimethylphenyl)amine)[12,15], carbazole-based self-assemble molecules (SAMs)[4,8] and nickel oxide (NiO_x)[6,16]. Unfortunately, these HTL materials still face challenges in combining high efficiency, stability and scalability in one device. Both poly(bis(4-phenyl)(2,4,6-trimethylphenyl)amine) and SAMs tend to have a non-wetting surface preventing uniform perovskite growth, which limits the scalability in large-area processing. NiO_x suffers from strong chemical reactions with perovskite at the interface due to its Lewis basicity and multivalent nature[17]. This interface redox reaction results in Ni^{3+} deprotonating perovskite and oxidizing halide ions, which causes an increase in non-radiative recombination at the interface and hence lowers the V_{oc} of the PSCs. To mitigate these issues, a bilayer HTL combining inorganic and organic materials has recently attracted many attentions. NiO_x/SAMs stand out as the typical bilayer HTL[6,18–20]. However, the spin-coated SAMs face another issue of forming a densely packed monolayer on NiO_x[21], which still causes recombination losses at this interface. Li et al. recently reported an inverted PSC with a PCE of 24.5% by using 1,3-bis(diphenylphosphino)propane to treat NiO_x/(4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid (Me-4PACz) and perovskite interface[6]. Overall, achieving a more efficient, conductive and stable interface material than NiO_x with high scalability remains challenging in the inverted PSC field.

As a prospective HTL material, SnO_2 has a lower standard electrode potential (<−0.1 V) compared to NiO_x, which indicates that Sn^{4+} is more difficult to be reduced. However, SnO_x-based HTLs have not been sufficiently explored in the inverted PSCs[22–24]. In particular, Sb-doped SnO_2 in which Sb^{3+} provides excess holes shows a potential to work as an excellent HTL material. Such a p-doping strategy can modify the SnO_2 to provide excellent hole conductivity while keeping its intrinsic stability at the same time[22]. It is important to emphasize that achieving p-type doping of SnO_2 with Sb^{3+} necessitates the avoidance of chlorine-based starting materials and high-temperature annealing processes (Supplementary Table 1). Furthermore, the Sb^{3+} is difficult to be oxidized as the standard electrode potential of Sb^{5+}/Sb^{3+} is +0.55 V.

Here we present antimony-doped tin oxide (ATO_x) combined with Me-4PACz as a HTL in efficient inverted PSCs. ATO_x effectively reduces efficiency losses in both small- and large-area inverted PSCs. Compared with NiO_x, the ATO_x layer forms a more chemically stable interface with perovskite. X-ray photoemission spectroscopy (XPS) analysis indicates that the redox reaction occurring at the NiO_x/perovskite interface is absent at the ATO_x/perovskite interface. Additionally, the ATO_x/Me-4PACz combination improves perovskite wettability, transmittance, conductivity and large-area uniformity. Photoluminescence quantum yields (PLQYs) and high-resolution external quantum efficiency (Hr-EQE) analysis reveal that ATO_x effectively suppresses non-radiative recombination in perovskites. Kelvin probe force microscopy (KPFM) at the nanoscale confirms that ATO_x enhances the depletion electric field at the perovskite/HTL interface. These combined improvements lead to an inverted PSC with maximum efficiency of 25.7% (certified steady-state efficiency of 24.8%) for an area of 0.05 cm^2 and 24.6% (certified steady-state efficiency of 24.0%) for an area of 1 cm^2 and stable performance under maximum power point tracking.

Characteristics of the ATO_x

To investigate the redox reaction between perovskite with Ni^{3+}, Sn^{4+} and Sb^{3+}, we first calculate the redox reaction energy between perovskite ($FAPbI_3$) and the corresponding metal oxides by density functional theory (DFT). The reaction energy between perovskite and Ni^{3+} is 1.0 eV. The value increases by a factor of one and reaches 2.1 eV for either Sn^{4+} or

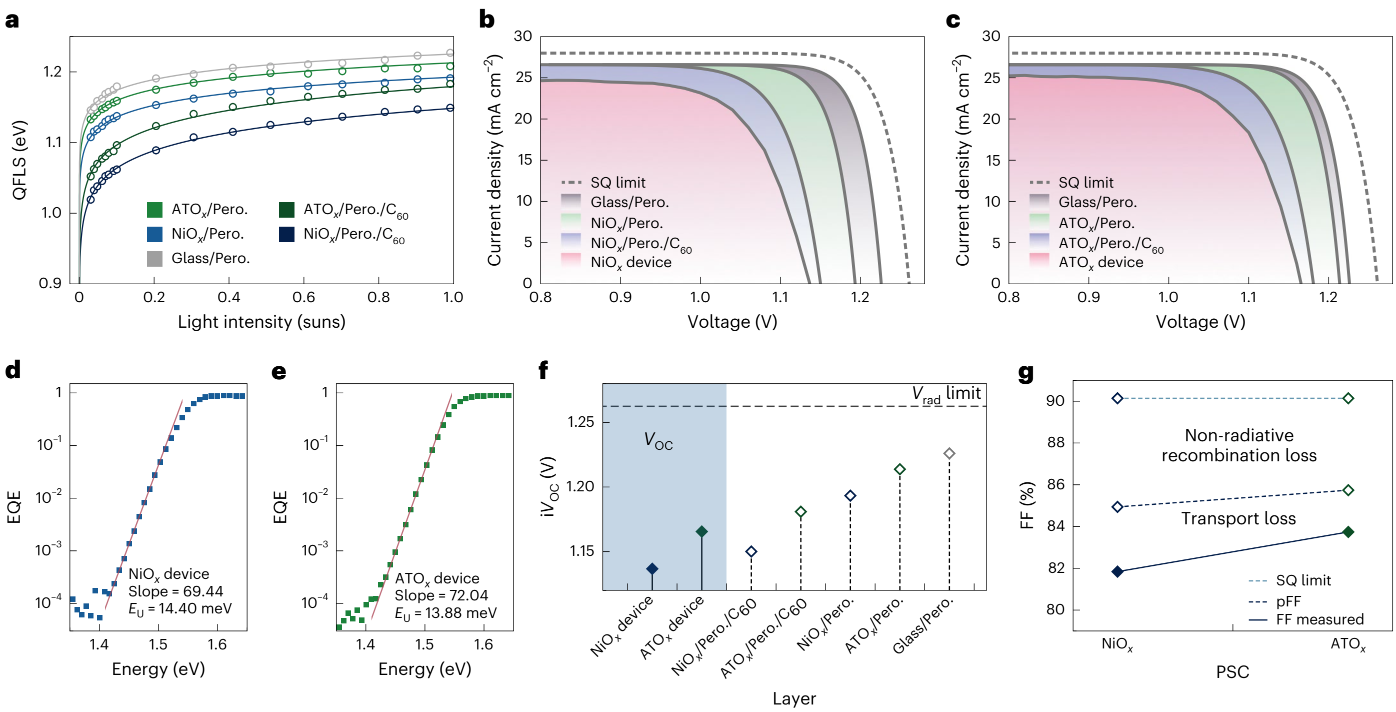

Fig. 2 | Non-radiative charge recombination analysis of ATO$_x$ PSCs and NiO$_x$ PSCs. a, Suns-QFLS plot of perovskite film and perovskite film with different interlayers under different light intensity illumination. The fitting curves are obtained from exponential fitting. **b**,**c**, Pseudo-*J*–*V* curves plotted from the light intensity vs QFLS. **d**,**e**, High-resolution EQE spectra of ATO$_x$ PSC (fitting function: $y = 4.05603\mathrm{e}-49 \times \exp(72.045 \times x)$) and NiO$_x$ PSC (fitting function: $y = 2.473695\mathrm{e}-47 \times \exp(69.4422 \times x)$). For the Urbach energy, $E_U = \frac{E-E_g}{\ln(\mathrm{EQE}(E))-\ln(\mathrm{EQE}(E_g))}$, where E_g is the bandgap of perovskite. By plotting ln(EQE(E)) against E, we can ascertain the Urbach energy, which is the inverse of the slope obtained from the linear fitting. **f**, V_{oc} loss analysis of perovskite on ATO$_x$ or NiO$_x$ and with other layers. The grey area shows the measured V_{oc} from the PSC devices, and the white area shows the calculated V_{oc} from the films. **g**, FF loss analysis of ATO$_x$ or NiO$_x$-based devices. The data points connected by the pale blue dashed line are ideal FF from the Shockley–Queisser (SQ) limit. The data points connected by the dark blue dashed line are the pseudo-FF from the pseudo-*J*–*V*. The data points connected by the solid line are the measured FF from the devices.

Sb^{3+} (Fig. 1a). This suggests that ATO$_x$ provides a more robust interface material against the redox reaction with perovskite.

ATO$_x$ nanoparticles are highly crystalline particles with a crystal size of about 10 nm, which is comparable to NiO$_x$ nanoparticles as reported previously (Supplementary Fig. 1a–c)[25]. The thickness of the ATO$_x$ layer deposited on ITO substrates was measured by atomic force microscopy (AFM), indicating a thickness of about 20 nm (Supplementary Fig. 1d). AFM images shown in Supplementary Fig. 2 further confirm that the morphology of the ATO$_x$ layer deposited on ITO substrates is more uniform than the NiO$_x$ layer. To compare the optical properties of ATO$_x$ and NiO$_x$, we measured their absorbances and transmittances using UV–vis spectroscopy. The ATO$_x$ shows higher transmittances from 300 nm to 900 nm (Fig. 1b) and a similar optical bandgap (4.46 eV) with that of NiO$_x$ (4.44 eV) (Supplementary Fig. 3). KPFM measurements reveal that both Me-4PACz-modified ATO$_x$ and NiO$_x$ share the same work function (Supplementary Fig. 4).

Although the same perovskite composition ($Cs_{0.1}FA_{0.9}PbI_3$) is used on both ATO$_x$ and NiO$_x$ substrates, the perovskite thin films on the two substrates exhibit distinct crystallinity and morphology. X-ray diffraction (XRD) measurement shows that the perovskite growth on the NiO$_x$ layer has more excess PbI_2 formation, which is the result of the redox reaction between perovskite and NiO$_x$ (Fig. 1c). Moreover, excess PbI_2 was also observed from the scanning electron microscopy (SEM) top-view images (Fig. 1d). XPS further confirms that the strong redox reaction between as-deposited NiO$_x$ and FAI at the interface during perovskite growth (Fig. 1e,f). The fittings of Ni 2*p* (Fig. 1e) and O 1*s* (Fig. 1f) spectral envelope from NiO$_x$ samples show the decrease of the $Ni^{\geq 3+}$/Ni^{2+} content ratio (Supplementary Table 2), which is mainly due to the redox reaction at the NiO$_x$/perovskite interface. In contrast, there is no obvious peak variation observed in the Sn 3*d* (Fig. 1g) and Sb 3*d* (Fig. 1h) spectral envelopes between as-deposited ATO$_x$ and ATO$_x$/FAI films, which further confirms the ATO$_x$ is robust and able to avoid reactions with perovskites. The redox reaction occurred at the interface between NiO$_x$ and the perovskite, leading to the formation of hole-trapping defects[17]. The dark current of the holes-only devices based on both NiO$_x$ and ATO$_x$ were measured (Supplementary Fig. 5). Notably, the NiO$_x$-based holes-only device exhibited a higher trap-filled limit voltage (V_{TFL}), indicating an increased trap density at the NiO$_x$ interface.

Characterization of the NiO$_x$- and ATO$_x$-based perovskite films

To investigate the influence of these HTLs on the device performance, in particular the device V_{oc} and FF, we carried out intensity-dependent steady-state photoluminescence (PL) measurements and recorded external PLQYs for different perovskite/HTL stacks (Supplementary Fig. 6). As shown in Fig. 2a–c, we plotted the calculated quasi-Fermi level splitting values (QFLSs) or implied V_{oc} (iV_{oc}) against the illumination intensity (Suns-V_{oc}) and then constructed pseudo-current density-voltage (pseudo-*J*–*V*) for both ATO$_x$ and NiO$_x$ stacks. The iV_{oc} and pseudo-FF values calculated from the pseudo-*J*–*V* curves are summarized in Supplementary Table 3. The neat perovskite films deposited on glass substrates exhibit the highest pseudo photovoltaic performance. After the addition of the transporting layers, interface recombination losses are leading to the decreased implied device performances[26]. In the perovskite/HTL stacks, the pseudo-PCE of the ATO$_x$-based stack reach 27.2% with an iV_{oc} of 1.21 V and pseudo-FF of 88.6%, which are higher than that of the NiO$_x$-based stack (pseudo-PCE of 26.3%, iV_{oc} of

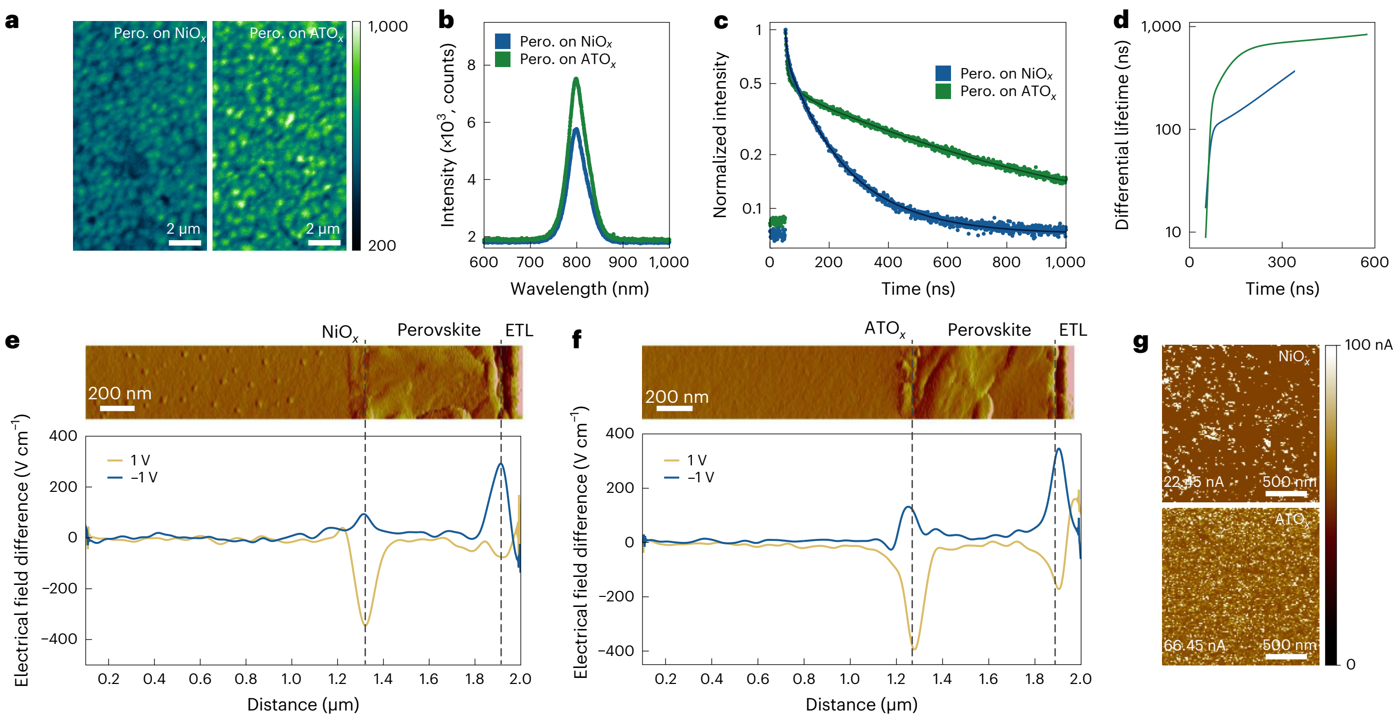

Fig. 3 | Interface charge-carrier dynamics and built-in electric field analysis. a, Confocal PL mapping images, the colour bar indicates the PL counts. **b**, Steady-state PL. **c**,**d**, TrPL and a differential lifetime of ATO_x films and NiO_x films. The black solid lines are the fitting to the corresponding TrPL profiles where a decay function with a tri-exponential model ($y0 + a1 \times \exp(-x/t1) + a2 \times \exp(-x/t2) + a3 \times \exp(-x/t3)$) is applied for the fitting. The fitting parameters have been included in Supplementary Table 4. **e**,**f**, The cross-sectional KPFM spectra and line profiles of electrical field difference under different bias voltage of 1 V and −1 V of NiO_x/perovskite sample (**e**) and ATO_x/perovskite sample (**f**). The vertical dashed lines indicate the interfaces between HTL/perovskite and perovskite/ETL. **g**, Conductive AFM images of NiO_x and ATO_x films, mean currents measured were 22.45 nA and 66.45 nA for NiO_x and ATO_x films, respectively.

1.19 V and pseudo-FF of 88.1%). In Fig. 2d,e, we measured Hr-EQE curves and then derived the radiative voltage limits to quantify voltage losses resulting from non-radiative recombination more precisely. We summarized the device V_{oc}, QFLS and radiative limit voltage values for both ATO_x and NiO_x devices/stacks in Fig. 2f. The iV_{oc} values derived for ATO_x stacks are consistently higher than for NiO_x stacks, which agree with the trend of measured V_{oc} in devices. In conclusion, perovskite growth on ATO_x suppresses the trap-induced non-radiative recombination and it translates to a reduction of non-radiative voltage losses from 0.12 V to 0.09 V. This is further verified by the Urbach energy (Eu) calculated from Hr-EQE. Moreover, the FF decrease caused by trap-induced non-radiative recombination is also mitigated, which leads to a 2% FF gain in the ATO_x device as shown in Fig. 2g.

To obtain deeper insights into the device performance improvement, we investigated the charge-carrier dynamics and the potential distribution at the perovskite/metal oxide HTL interface. Steady-state PL spectra illustrate that the ATO_x/Me-4PACz/perovskite film can achieve higher absolute PL intensity than NiO_x/Me-4PACz/perovskite under the same measuring conditions (Fig. 3b). Similarly, the confocal PL images shown in Fig. 3a further verify the high emission yield and uniformity of the perovskite film deposited on ATO_x substrate. Time-resolved photoluminescence (TrPL) measurement was conducted to analyse the holes transferring from perovskite into the HTLs and trap-assisted charge recombination (Fig. 3c). For direct reading of the transient decay time, we plotted the differential lifetime according to the equation, $\tau = -(\mathrm{d}\ln(\phi(t))/\mathrm{d}t)^{-1}$ (Fig. 3d and Supplementary Table 4). The rise of τ at early times for ATO_x-based perovskite film is clearly faster than for NiO_x/perovskite stack, indicating ATO_x HTL is beneficial for efficient charge extraction[8]. Furthermore, according to the fitting of the monoexponentially decay at later times, ATO_x/Me-4PACz/perovskite exhibits a substantially longer carrier lifetime compared with NiO_x/ Me-4PACz/perovskite, demonstrating an effective suppression of non-radiative recombination for the perovskites deposited on ATO_x.

To investigate the influence of different HTLs on the electrical field in the device, we used cross-sectional KPFM to profile and observe the electric-field distribution in the vertical direction. (Fig. 3e,f). Different bias voltages (from −1 V to 1 V) were applied during the measurement to minimize the influence of surface charges (Fig. 3e,f and Supplementary Fig. 7). We then derived the electric field by taking the first derivative of the net potential change resulting from the bias voltages. By aligning the profiles to the topography, we were able to determine the electric field distribution within the device. When a bias voltage of 1 V is applied, a prominent peak emerges at the location of the perovskite/HTL interface for both NiO_x and ATO_x, whereas the electric field was near zero inside the perovskite layer. The electric field difference demonstrates a statistically higher value for devices based on ATO_x, as indicated in Supplementary Fig. 8. This elevated performance is attributed to ATO_x's ability to suppress surface reactions and minimize defect formation in the perovskite layer. Another depletion at the perovskite/C_{60} interface on ATO_x was also enhanced. Moreover, the conductive AFM (c-AFM) measurements (Fig. 3g) show that ATO_x (66.45 nA) film is more conductive than NiO_x film (22.45 nA). As a result, the combination of stronger electric fields with more conductive transporting layers leads to efficient charge extraction in ATO_x PSCs.

Performance of the NiO_x- and ATO_x-based devices

Both NiO_x PSCs and ATO_x PSCs are based on a p–i–n configuration consisting of glass/indium-doped tin oxide (ITO) or fluorine-doped tin oxide (FTO)/NiO_x or ATO_x/Me-4PACz/$Cs_{0.1}FA_{0.9}PbI_3$/C_{60}/bathocuproine

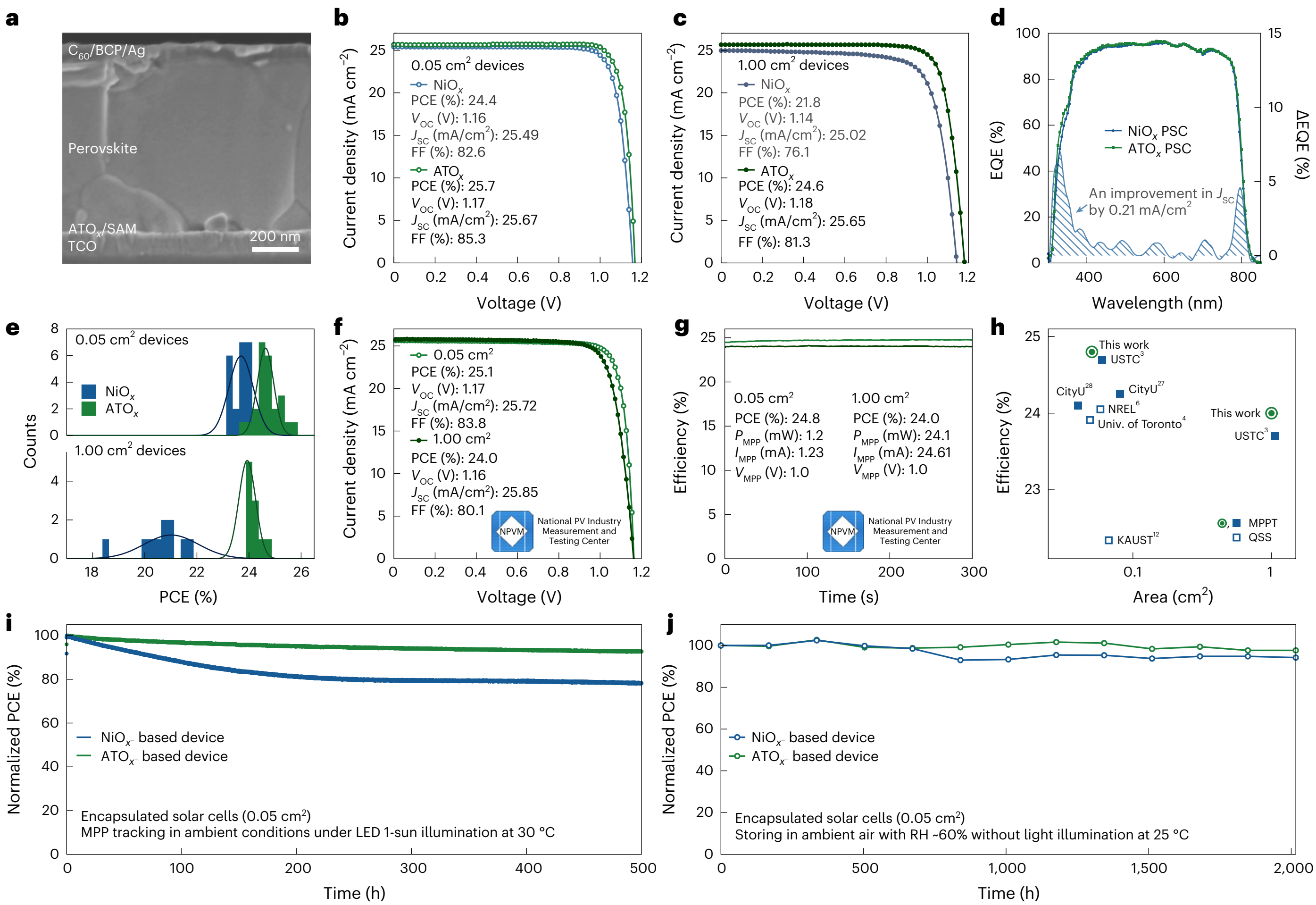

Fig. 4 | Device characteristics. a, Cross-sectional SEM image of the typical inverted PSCs. **b**,**c**, *J*–*V* curves of 0.05 cm^2 (**b**) and 1 cm^2 (**c**) champion NiO_x PSC and ATO_x PSC. **d**, EQE profiles of NiO_x PSC and ATO_x PSC. Difference between the EQE values of the two devices is calculated and drawn with blue curve filled with blue dashes underneath (*y* axis on the right-hand side). Overall, an improvement of 0.21 mA cm^{-2} is found from the ATO_x PSC. **e**, PCE distribution of 0.05 cm^2 PSCs (24 devices for both NiO_x PSCs and ATO_x PSCs) and 1 cm^2 PSCs (ten devices for both NiO_x PSCs and ATO_x PSCs), the distribution is fitted with Gaussian function. **f**,**g**, *J*–*V* (**f**) and MPPT measurements (**g**) of 1 cm^2 device recorded by NPVM. **h**, Comparison of PCE between this work and other reported steady-state PCEs of *p*–*i*–*n* PSCs[3,4,11,16,30,31]. The devices certified by MPPT and quasi-steady-state (QSS) measurement are denoted with solid and hollow squares, respectively. Institutes included: City University of Hong Kong (CityU), King Abdullah University of Science and Technology (KAUST), National Renewable Energy Laboratory (NREL), University of Science and Technology of China (USTC), University of Toronto (Univ. of Toronto). The devices in this work are certified by MPPT. **i**, Operational stability under 1-sun illumination in ambient air with a relative humidity of ~60% and a fan was used to cool the device to room temperature (~25 °C), initial PCEs are 23.7% and 25.0% for encapsulated NiO_x PSC and ATO_x PSC, respectively. **j**, Shelf stability in 60% relative humidity, dark environment at room temperature (~25 °C), initial PCEs are 22.3% and 24.3% for encapsulated NiO_x PSC and ATO_x PSC, respectively. Credit: **f**,**g**, National PV Industry Measurement and Testing Center.

(BCP)/Ag as shown by the cross-sectional SEM in Fig. 4a. The performance evaluation of ATO_x PSCs was conducted using varying concentrations of antimony-doped SnO_2 (Supplementary Fig. 9). On the basis of the outcomes, we selected a 10% doping concentration as the optimized ATO_x material. This optimized ATO_x layer was chosen for comparison with NiO_x in our investigation. The 0.05 cm^2 ATO_x PSC delivered a maximum PCE of 25.7%, showing a relatively small efficiency gap of 1.3% compared with NiO_x PSC (Fig. 4b). In strong contrast, such PCE gap increased to 2.8% in 1 cm^2 device (Fig. 4c). Both J_{sc} and V_{oc} of ATO_x PSC were nearly unchanged when the device area was increased from 0.05 cm^2 to 1 cm^2. The slight drop of FF is mainly induced by the series resistance from substrates. This result is significantly different from NiO_x PSC whose PCE decreased from 24.6% to 21.8%. This result is also in line with the PCE statistics (Fig. 4e and Supplementary Fig. 10). Moreover, due to the enhanced transmittance of the ATO_x layer (Fig. 1b), we observed an enhanced EQE in ATO_x PSCs from 300 nm to 800 nm (Fig. 4d). As a result, it leads to a consistent J_{sc} improvement of ~0.2 mA cm^{-2} in ATO_x PSC compared with NiO_x PSC. (Fig. 4d).

By improving the conductivity and uniformity of large-area ATO_x deposition (Fig. 3g), we achieved the National PV Industry Measurement and Testing Center (NPVM)-certified stabilized PCE of 24.8% (0.05 cm^2) and 24.0% (1 cm^2) (Figs. 4f,g and Supplementary Figs. 11 and 12) under a maximum power point tracking (MPPT) over 300 seconds duration. Comparing our certified stabilized PCE results with the state-of-the-art devices (Fig. 4h), our ATO_x PSCs exhibit one of the highest PCEs in 1-cm^2 area and possess advantages for upscale manufacturing compared with Me-4PACz-only PSCs (Supplementary Figs. 13 and 14).

The devices reported herein also show excellent operational stability under accelerated tests (Figs. 4i,j and Supplementary Fig. 15). We monitored the stability of both encapsulated and unencapsulated devices. The encapsulated devices (ATO_x PSC and NiO_x PSC) were stored in ~60% relative humidity environment and both can maintain above 90% of the initial PCE (Supplementary Fig. 16) performances for 2,000 hours. We also monitored the 500-hour operating stability of encapsulated devices at 30 °C with ~70% relative humidity. We found that ATO_x

PSC can maintain above 93% of the initial PCE (Supplementary Fig. 17) after 500 hours operating under 1-sun conditions whereas NiO_x PSC can maintain 77% only.

Methods

Materials

Caesium iodide (CsI, 99.99%) lead chloride ($PbCl_2$), dimethylformamide (DMF, anhydrous) and 1-Methyl-2-pyrrolidinone (NMP, anhydrous) were purchased from Sigma-Aldrich. Formamidinium iodide (FAI) was purchased from Greatcell Solar Materials; anhydrous lead iodide (PbI_2) and (4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl)phosphonic acid (Me-4PACz) were purchased from Tokyo Chemical Industry. Silver was purchased from Kurt J. Lesker, and C_{60} and BCP were purchased from Lumtec. ITO substrates (X07-20AC) or FTO substrates (A11-UV) were purchased from Suzhou Shangyang Solar Technology Co. The ATO_x suspension (#10167) was supplied by Avantama AG.

NiO_x nanoparticles were prepared following previously published report[27]. First, nickel(II) nitrate hexahydrate (9 g) was fully dissolved in 120 ml deionized water, in which 120 ml sodium hydroxide (1 mol ml^{-1}) was added under vigorous stirring at room temperature. The solution was stirred for another 5 min to complete the reaction; then green raw products ($Ni(OH)_2$) were collected by centrifugation at 3,600 × g for 8 min, followed by thorough washing with deionized water. The above centrifugation and washing steps were repeated at least three times until the pH values of the supernatants were neutral. Then, the obtained raw products were thoroughly dried with freeze-drying equipment for at least 48 h. Finally, the powders were sintered at 275–280 °C for 2 h with a tube furnace. The resulting NiO_x powders appear to be black in colour and are dispersed in deionized water to form nanoparticle ink (10 mg ml^{-1}).

ATO_x nanoparticles (#10167) were produced through flame spray pyrolysis by Avantama AG[22]. First, antimony(III) acetate and tin(II) 2-ethylhexanoate precursors were mixed in a targeted molar ratio. In this case, 10% of antimony(III) acetate was added. The precursor mixture was then dissolved in a mixture of alkyl carboxylic and toluene. The precursor solution was then fed into a spray nozzle at a flow rate of 9 ml min^{-1}, dispersed by oxygen at a flow rate of 14 l min^{-1} and ignited by a methane-oxygen premixed flame (CH_4, 1.2 l min^{-1}; O_2, 2.2 l min^{-1}). The off gas was filtered by a vacuum pump through a mesh filter at approximately 20 m^3 h^{-1}. The resulting ATO_x powders were collected from the mesh filter and subsequently dispersed in a butanol mixture at a concentration of 2.5% by weight.

Computational details

All the spin-polarized DFT calculation was performed in the plane-wave pseudopotential method based on the generalized gradient approximation-perdew-burke-ernzerhof (GGA-PBE) exchange-correlation functional[28], which is implemented by Vienna ab initio simulation package (VASP, 5.4.4 version[29]). The valence wave functions were expanded by plane wave with a cut-off energy of 400 eV. We established the model of involved $FAPbI_3$ (tetragonal phase, space group: I4/mcm), metal oxides and metal hydroxides using their primitive cell with 7 × 7 × 7 Monkhorst–Pack k-point sampling mesh. For the gas molecules, we established a 15 × 15 × 15 Å unit cell to calculate the energy of isolated gas molecules. The Brillouin zone was sampled using only the zone centre Γ for molecular geometry optimization. All the structures were optimized until the force on each atom was less than 0.02 eV $Å^{-1}$.

Device fabrication

The patterned ITO or FTO glass substrates were wiped with soap water, then ultra-sonically washed with deionized water, acetone and isopropanol for 30 minutes each. After ultraviolet ozone treatment for 15 min, the substrates were then spin coated with ATO_x (2.5 wt% in butanol) or NiO_x (10 mg ml^{-1} in water) nanoparticles, followed by heating at 100 °C for 10 minutes in the open air. Then the Me-4PACz (1 mg ml^{-1} in IPA) was spin coated on the substrates at 4,000 r.p.m. for 30 s and heated at 100 °C for 10 min in the nitrogen-filled glovebox. The 1.8 M perovskite ($Cs_{0.1}FA_{0.9}PbI_3$) stock solution contains 52 mg CsI, 309.6 mg FAI, 922 mg PbI_2 and 55.6 mg $PbCl_2$ that were dissolved in 1.11 ml mixed solvent of DMF: NMP (V:V = 960:150). The 1.8 M perovskite precursor solution was shaken overnight to make it fully dissolved and then used for preparing perovskite films. For the spin-coating process, the substrate was spinning at 5,000 r.p.m. for 40 s with an acceleration of 5,000 r.p.m. s^{-1} and N_2 gas was blown on top of the spinning substrates after 20 s. The perovskite films were pre-annealed at 70 °C for 2 minutes in the N_2 glovebox to eliminate any residual solvent and then annealed at 150 °C for 10 minutes in 30% relative humidity to promote the crystallinity of perovskite with enhanced film quality and grain size. Lastly, C_{60} (20 nm)/BCP (8 nm)/Ag (100 nm) were deposited to complete the device fabrication. The devices were masked with metal aperture masks (0.05 cm^2 and 1 cm^2) during the J–V measurement.

Device characterization

J–V measurements of the devices were recorded with a Keithley 2400 source meter under simulated 1-sun AM 1.5 G illumination (100 mW cm^{-2}) with an ABET Technologies Sun 2000 solar simulator. The measurements are performed in an N_2-filled glovebox. The illumination area of the devices is defined by shadow masks of 0.05 cm^2 and 1.00 cm^2 area. MPPT measurements of the devices were recorded with MPP Tracking–4B system (Shenzhen Lancheng Technology Ltd.) with light-emitting diode (LED) simulated AM 1.5 G spectrum. The measurements were carried out at ambient conditions at 25 °C with relative humidity (RH) of ~60% with the devices being encapsulated. EQE measurements were conducted using a Bentham PVE300-IVT system. The LED laser intensity is calibrated with built-in silicon and germanium diodes before the measurements. For the high-resolution EQE, the light is chopped at 137 Hz and coupled into a Bentham monochromator. The resulting monochromatic light was focused onto the perovskite solar cell, and its current under short-circuit conditions was fed to a current preamplifier (Stanford SR 570) before it was analysed with a lock-in amplifier (Stanford SR830 DSP). The time constant of the lock-in amplifier was chosen to be 1 s and the amplification of the preamplifier was increased to resolve low photocurrents. The EQE was determined by dividing the photocurrent of the cell by the flux of incoming photons, which was measured using a calibrated Si photodiode. All EQE measurements have been performed at ambient conditions at 25 °C and with RH of around 60% and without any encapsulation. Suns-V_{oc}, PLQY and electroluminescence measurements were conducted with LP20-32 radiative efficiency meter (QYB). For suns-V_{oc} measurements, the laser was switched on 15 min before the measurement to allow it to warm up and stabilize. The laser is then set to desired intensities and performs PL measurement. Plotting of the suns-V_{oc} and pseudo-J–V follows previous reports[8]. For the pseudo-J–V, 95% of SQ current density is used to avoid potential influences for the iV_{oc} comparison.

Material characterizations

SEM images were taken with Regulus SU8200 system (Hitachi) at 3 kV accelerating voltage under SE mode. TEM measurements were conducted with JEM-2010 system (JEOL). Absorbance and transmittance spectra were recorded with Agilent Cary 7000 spectrophotometer. The system is allowed for 20 min to warm up the lamp before the measurement. NiO_x and ATO_x nanoparticle dispersions of different concentrations were prepared. The baseline was first calibrated with their corresponding solvents, followed by the measurements of the liquid dispersion samples. The transmittance spectra were then gathered and analysed to obtain the materials' optical bandgap. XRD is recorded on an X-ray diffractometer (Malvern Panalytical) with Cu Kα radiation (λ = 0.1542 nm). XPS measurements were performed with Kratos Axis Ultra XPS system. Samples were stored under high vacuum conditions

https://doi.org/10.1038/s41560-023-01442-1

overnight to minimize influences from the absorbed H_2O and other particles on the surface. Peak fitting and deconvolution followed previous reports[17]. A Shirley-type background was applied and a Gaussian–Lorentzian convolution, GL(30), was used for deconvoluting the peaks. The peak position of NiO was set to ~854.13 eV, following with $Ni(OH)_2$/Ni_2O_3 (+0.85 eV), NiO(OH) (+2.00 eV) and $Ni^{\geq 3+}$ (+3.25 eV) were used for fitting the Ni $2p_{3/2}$ spectra. Fitting parameters and processes for the rest of the spectra remain similar. 0.1 M FAI solution was utilized and spin coated at 5,000 r.p.m. for 30 s to ensure that the resulted film thickness is compatible with the XPS detection range. TrPL measurements were conducted by PicoQuant FluoTime 300 spectrometer. The area of interest was first found and located under the optical microscope. After tuning the lens to their optimum position, the steady-state PL and the lifetime measurement are conducted with laser excitation at 515.8 nm in ambient air with the films encapsulated. Confocal PL mapping was measured on a Nikon A1 confocal microscope equipped with a 633 nm continuous-wave laser. AFM, KPFM and c-AFM measurements were conducted with the Park NX20 system. AFM measurements were performed with non-contact mode. Sideband KPFM measurements were performed under EFM (non-contact) mode. The work function was calibrated before every sample with a standard gold reference sample. Also, the signals were optimized in advance by applying bias voltage onto the samples in advance of the measurements. The c-AFM measurements were performed under contact mode. The cantilever tip bias was constantly checked before and during the measurements and the current amplifier was tuned and optimized before the c-AFM measurements. Cross-sectional KPFM experiments were performed in a custom-built system located in an Ar-filled glovebox. To expose the cross-sectional surface, the devices were cleaved from the film side. We applied bias voltages ranging from −1 V to +1 V to the device during measurements while scanning the same area. For data processing, we averaged the potential profiles and subtracted the 0-V profile to eliminate any surface effect. To derive the electric field, we took the first derivative of the net potential change caused by the bias voltages. By aligning the profiles with the topography, we determined the distribution of the electric field within the device.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request. Source data are provided with this paper.

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Acknowledgements

Y.H. acknowledges the support from MOE Tier 2 grant (MOE-T2EP10122-0005), the Ministry of Education (Singapore) and the National University of Singapore Presidential Young Professorship (A-0009174-03-00 and A-0009174-02-00). This research is supported by the National Research Foundation, Singapore and A*STAR (Agency for Science, Technology and Research) under its LCERFI programme award number U2102d2002. We are affiliated with the Solar Energy Research Institute of Singapore (SERIS), a research institute at the National University of Singapore. SERIS is supported by the National University of Singapore, the National Research Foundation Singapore, the Energy Market Authority of Singapore and the Singapore Economic Development Board. We also acknowledge generous computing resources at National Supercomputing Centre (NSCC) Singapore.

Author contributions

Y.H., J.L. and H.L. conceived the idea and designed the experiments. Y.H. directed and supervised the project. J.L. fabricated the perovskite solar cells. H.L. conducted material and solar cell characterizations. C.X. assisted with cross-sectional KPFM measurements. X.J. conducted high-resolution EQE measurement and assisted with V_{oc} loss analysis. X.G. and J.F. assisted with confocal PL measurements. J.C. and R.L. assisted with reaction energy calculation. H.L., X.J and R.G. carried out the TrPL measurements and analysed the data. X.W. assisted with device fabrication and measurements. M.L. assisted with cross-sectional KPFM measurements and data analysis. A.H., F.L. synthesized the ATO_x nanoparticles, A.H. produced the ATO_x formulations. M.R. oversaw the ATO_x production process. J.L., H.L., E.A., S.L. and Y.H. analysed data and wrote the manuscript. All authors read and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41560-023-01442-1.

Correspondence and requests for materials should be addressed to Yi Hou.

Peer review information *Nature Energy* thanks Xingwang Zhang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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Corresponding author(s): Yi Hou

Solar Cells Reporting Summary

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Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

	Yes	No	
Area of the tested solar cells	☒	☐	The area of tested solar cells is 0.05 cm2.
Method used to determine the device area	☒	☐	Metal aperture mask (black color)

2. Current-voltage characterization

	Yes	No	
Current density-voltage (J-V) plots in both forward and backward direction	☒	☐	We provide J-V plots in both forward and backward direction in Supplementary Fig. 8 and 9.
Voltage scan conditions *For instance: scan direction, speed, dwell times*	☒	☐	See details in Method.
Test environment *For instance: characterization temperature, in air or in glove box*	☒	☐	See details in Method.
Protocol for preconditioning of the device before its characterization	☐	☒	No preconditioning condition is used.
Stability of the J-V characteristic *Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.*	☒	☐	We provided MPPT data in Fig.4G

3. Hysteresis or any other unusual behaviour

	Yes	No	
Description of the unusual behaviour observed during the characterization	☐	☒	Negligible hysteresis was observed.
Related experimental data	☐	☒	Hysteresis is not relevant to our concept.

4. Efficiency

	Yes	No	
External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒	☐	We provide EQE measurement in Fig. 4D
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒	☐	We compare the integrated Jsc with one from JV scan. The difference between the integrated Jsc from EQE and Jsc from JV scan is within 1% difference, which is within accuracy confidence of the measurements.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐	☒	We didn't fabricate tandem solar cell.

5. Calibration

	Yes	No	
Light source and reference cell or sensor used for the characterization	☒	☐	See details in Method.
Confirmation that the reference cell was calibrated and certified	☒	☐	Our solar simulator was calibrated by Si reference cell.(certificated by Newport.)

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Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	The champion devices were sent to do certification, the certified device spectra mismatch factor (1.008 and 1.001) can be found in Supplementary Fig. 8 and 9.

6. Mask/aperture

Size of the mask/aperture used during testing	☒ Yes ☐ No	An optical aperture mask (0.05 cm2) was used.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	No significant variations was observed.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	We provide certification data in Supplementary Fig. 8 and 9.
A copy of any certificate(s) *Provide in Supplementary Information*	☒ Yes ☐ No	We provide certification data in Supplementary Fig. 8 and 9.

8. Statistics

Number of solar cells tested	☒ Yes ☐ No	See details in Fig. 4E caption
Statistical analysis of the device performance	☒ Yes ☐ No	We provide statistics in Fig. 4E

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions *For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature*	☒ Yes ☐ No	We provided stability data in Fig.4I and J